

Fahim Ahamed

Authorized to work in the U.S. and do not require sponsorship - U.S. Permanent Resident

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Summary

Data Scientist with a strong background in **AI Research**, with experience across **healthcare**, **NLP**, and **geoscience** domains. Builds **end-to-end ML pipelines**, conducts **statistical and spatial analysis**, and deploys models that hold up in production. Develops **agentic AI pipelines** using **RAG**, **LLMs**, and **MCP**. Published **peer-reviewed research** across **IEEE**, **SAGE**, and **Springer Nature**. Comfortable working across the full stack from raw data to results, and collaborating directly with domain experts. Proficient in **Python**, **SQL**, **R**, **PyTorch**, and **TensorFlow**.

Education

The City College of New York

New York, NY

M.S. in Data Science and Engineering | GPA: 3.83/4.0

Dec 2027

East West University

Dhaka, Bangladesh

B.S. in Computer Science and Engineering | GPA: 3.74/4.0

Dec 2023

Technical Skills

Languages: Python, SQL, R

Data Analysis: NumPy, Pandas, SciPy, Excel

Data Visualization: Matplotlib, Seaborn, Plotly, ggplot2, Tableau

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, NLTK, Word2Vec

LLMs & Agents: LLMs, RAG, MCP, Hugging Face, ChromaDB

Database & Big Data: MySQL, MongoDB, Snowflake

MLOps & Cloud: AWS, MLflow, Git/GitHub, Docker, Vercel

Work Experience

Data Analyst

Oct 2025 - Feb 2026, Jul 2026 - Present

The City College of New York

New York, NY

- Analyze **institutional data** using **Excel**, **Tableau**, and **Python**, surfacing patterns that shape **advising and student engagement programs**.
- Conduct **data-driven surveys** and **retention studies**, uncovering behavioral patterns that inform targeted outreach.
- Build **analytical reports** and **dashboards** that surface trends and anomalies across departments, replacing gut-feel decisions with **data-backed ones**.

Research Assistant

Dec 2025 - Present

CUNY Research Foundation

New York, NY

- Develop an **agentic AI pipeline** using **RAG**, **LLMs**, and **MCP-connected tools** for natural language querying over complex geoscience databases, as part of an **NSF-funded** geoinformatics project.
- Bridge **geoscience domain expertise** and **AI engineering**, translating research requirements into technical specs and iterating on pipeline outputs for interpretability and scientific relevance.
- Design the **end-to-end technical stack: vector database architecture**, domain-specific **embedding models**, **LLM optimization**, and **cloud deployment** for a production-ready geoscience research tool.

Data Evangelist Intern

Feb 2026 - May 2026

DataCamp

New York, NY

- Designed and maintained **scalable data pipelines** to collect, integrate, and validate **multi-source data** through automated scripts, APIs, and web scraping.
- Built **data applications** and **analytical workflows** powered by **LLMs**, integrating **OpenAI** and **Hugging Face** models into user-facing tools for exploring key metrics.
- Analyzed data to surface trends and key drivers, packaging findings into **visualizations** and structured reports.

Research Experience

Explainable AI-Driven Hybrid Transfer Learning Framework for Accurate Skin Cancer Diagnosis – *Digital Health (SAGE)*

- Achieved **98.65% accuracy (F1 98.64%)** on multi-class skin cancer classification using a hybrid EfficientNetB0 + Random Forest model with advanced preprocessing, validated across datasets.

Robust Dual-Site Cancer Screening via Multi-Scale Vision Transformer and Rapid Recognition Pipeline – *IEEE Xplore*

- Developed a **multi-scale Vision Transformer framework** (Multi-ViT, MA-Transformer V2, MobileUNETR, TinyViT, Xception) achieving **99.12% accuracy** on PAD-UFES-20 and **99.37%** on Oral Cancer datasets, with strong F1 scores.

Texture Feature-Based Colonic Polyp Detection Using Deep Learning Techniques – *Springer Nature*

- Achieved **97.33% IoU** for colonic polyp segmentation using DeepLabv3+, outperforming VGG16 and prior studies to set a new benchmark in colorectal cancer detection.

Optimizing Energy Usage: Genetic Algorithms Leading the Charge Toward Sustainability – *Springer Nature*

- Reduced** annual household electricity usage by **34.54%** (90,449 watts) using **Genetic Algorithms**, demonstrating strong environmental impact and potential for **nationwide scalability** in sustainable energy optimization.

Accepted Papers (In Press)

A Lightweight Transformer Ensemble for Explainable Depression Emotion and Severity Detection – *IEEE (2025)*

- Achieved **state-of-the-art F1 81.63%, Precision 83.24%, Specificity 86.92%** on benchmark datasets with a lightweight transformer ensemble, providing **LIME-based explanations** and a deployable real-time interface for depression detection.

Data Projects

NYC HIV/AIDS Trend Analysis & Predictive Modeling | *R, Tidyverse, ggplot2, sf, rpart, RandomForest, caret*

- Cleaned and engineered multi-year NYC Open Data on HIV/AIDS diagnoses, resolving **suppressed values** withheld for privacy, into analysis-ready datasets for modeling.
- Performed **spatiotemporal analysis** across NYC, uncovering persistent geographic and demographic disparities.
- Applied **Random Forest models ($R^2 = 0.65$)** to identify high-impact predictors for public health targeting.

Global Population Growth Modeling & Forecasting | *R, Tidyverse, ggplot2, leaps, caret*

- Engineered a **longitudinal time-series dataset** from World Bank indicators across **200+ countries** and **60+ years** of demographic trends.
- Conducted multi-decade demographic analysis by region and income group, revealing **divergent growth trajectories** between developing and developed economies.
- Developed forecasting models achieving **$R^2 = 0.755$** using **rolling-window cross-validation**.

NYC 911 Mental Health Dispatch Analysis | *Python, Pandas, GeoPandas, Plotly, HTML/CSS, JavaScript*

- Built a **reproducible ETL pipeline** over **2.59M NYC 911 dispatch records**, deduplicating to a clean crisis-incident dataset and joining census blocks to precincts.
- Applied **per-capita normalization** and **temporal decomposition** to quantify crisis burden across **78 precincts**, separating a real decline from a shift in dispatch practice.
- Published the analysis as an interactive **Plotly** visualization, surfacing precinct-level and seasonal patterns for a non-technical audience.

AI/ML Projects

Cursive Handwritten Prescription Recognition | *Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, Matplotlib*

- Designed a **custom CNN architecture** to digitize mixed Bangla-English prescriptions, handling cursive and multilingual handwriting.
- Built **modular preprocessing pipelines** for noise reduction, segmentation, and character recognition, improving scalability across varied handwriting styles.
- Achieved **91% OCR accuracy** and reduced training time by **20%** through optimized hyperparameter tuning.

Intrusion Detection System | *Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*

- Preprocessed **2.8M+ network traffic records** across 7 attack types and 78 features, resolving missing values and downcasting numeric dtypes to reduce memory usage by **47.5%**.
- Applied **incremental PCA** for dimensionality reduction and **SMOTE** for class imbalance, then benchmarked five models across binary and multi-class detection tasks.
- Achieved **97% median accuracy** in anomaly detection, with **SVM** leading binary and **Random Forest** leading multi-class classification.